

- Prompts: reusable prompt templates that standardise model interactions. Demonstrated with an industry example
- Walkthrough: how a single MCP server can expose all three primitives simultaneously
- Key design principle: why descriptions and schemas matter for model understanding

Module C: MCP in Industry & the Ecosystem (20 min)

- Real-world MCP server showcases from the McpToTheMoon community across FinTech, HealthTech, and Manufacturing
- Agentic patterns: tool chaining, multi-step reasoning, and human-in-the-loop workflows
- Widgets and rich UI: how MCP outputs can be surfaced in chat interfaces and embedded applications
- Security and authentication patterns for production MCP servers
- Overview of what teams will build during the hackathon and what good looks like

SESSION 3 | 10:30 – 11:30 AM

Nitrostack Product Demo & Platform Walkthrough

Presenter: Nitrostack Team | Format: Live demonstration | Duration: 60 minutes

Part 1: Nitrostack Suite Introduction (10 min)

- Overview of Nitrostack: Studio, CLI, SDK, and NitroChat
- The Nitrostack philosophy: from idea to deployed MCP server with minimal friction
- Supported languages, runtimes, and platform integrations

Part 2: CLI Introduction (10 min)

- Installing Nitrostack: ``npm install -g nitrostack``
- Project initialisation: ``nitro init my-mcp-server``
- Core developer commands: ``nitro dev`` (local), ``nitro build``, ``nitro deploy``
- Secrets and environment variable management

Part 3: Studio Walkthrough (15 min)

- Opening a project in Nitrostack Studio: the visual development environment for MCP
- Tool editor: defining tool inputs, outputs, and natural language descriptions
- Resource browser: connecting external databases, APIs, and file sources
- Prompt designer: building and versioning reusable prompt templates
- Live testing panel: running tool calls against your MCP server without leaving the IDE

Part 4: AI-Assisted Development Walkthrough (10 min)

- AI-assisted development inside Studio: generate tool stubs from plain-English descriptions
- Iterating with natural language prompts to modify tool behaviour
- Auto-generated schema validation and documentation

Part 5: Deployment Walkthrough (15 min)

- One-command deployment: ``nitro deploy --prod``
- ChatGPT App deployment: exposing your MCP server as a GPT plugin
- NitroChat App deployment: embedding your server in a branded conversational UI
- Monitoring and observability: request logs, latency, and error dashboards
- Hackathon submission portal: a live walkthrough of what teams will see and how to submit their project

SESSION 4 | 11:30 AM – 12:00 PM

Hackathon Announcement, Rules, Registration & Kickoff

The rules are read aloud and displayed on the main screen. Pre-registered and accepted teams proceed to check-in and confirmation. The hackathon begins at 12:00 PM sharp.

PARALLEL SESSION | 02:00 – 04:00 PM

Faculty Session: Use of AI by Educators

Audience: Faculty members of Amrita University | Format: Talk + Discussion | Duration: 2 hours

Venue: Separate seminar room / hall (running parallel to the student hackathon)

Session Overview

By agreement with Amrita University, a dedicated 2-hour session for faculty will run in parallel with the hackathon. This session is designed to help educators understand how AI, and specifically Agentic AI and MCP, can be practically applied in teaching, assessment, curriculum design, and research.

Proposed Topics

- How Large Language Models work: a practical, non-technical primer for educators
- AI in the classroom: use cases across disciplines (Engineering, Management, Life Sciences, Architecture)
- Using AI for curriculum design, lesson planning, and personalised learning paths
- Assessment in the age of AI: rethinking evaluation, plagiarism, and academic integrity

- Research acceleration: how AI tools assist literature review, data analysis, and hypothesis generation
- Agentic AI and MCP in education: a look at what students are building today, and what it means for the future of learning
- Hands-on demonstration: a faculty-focused MCP server connecting an AI agent to institutional knowledge bases
- Open discussion: challenges, concerns, and opportunities for educators

Faculty Session Note

This session is not part of the competitive hackathon. Faculty attendance is by invitation from Amrita University. The session is intended to be conversational and exploratory: no prior AI knowledge is required. Facilitators will be drawn from the Nitrostack team and invited educators with experience in AI-assisted pedagogy.

4. Hackathon Rules & Guidelines

These rules apply to all participants from the moment of registration. Violation of any rule may result in disqualification. All decisions by the organising team and judging panel are final.

4.1 Eligibility

R 1	Open to all currently enrolled students of Amrita University (UG, PG, and PhD programmes). Faculty members may attend the morning sessions and mentor teams but are not eligible to compete.
R 2	Each participant may be registered with only one team. Dual registration across multiple teams is not permitted and will result in disqualification of all teams involved.
R 3	Valid student identification is required at on-campus check-in on 17 July 2026. Teams will be verified against the current enrolment records.
R 4	At least one team member is expected to be physically present at the venue. This requirement may be waived at the discretion of the organising team depending on final venue capacity. The organising committee will communicate the final policy to all registered teams before the event day.

4.2 Team Formation

R 5	Teams must consist of 2 to 4 members. Solo participation is not permitted.
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R 6	Cross-disciplinary teams are strongly encouraged. The strongest submissions typically combine technical depth (CS, ECE, IT) with domain expertise (Management, Life Sciences, Architecture, Law). Teams are rewarded for this in the judging rubric under Real-World Impact.
R 7	Teams must register online via the Nitrostack portal before the published registration deadline (prior to 17 July 2026). Registrations are validated and accepted by the organising team. Accepted teams receive a confirmation. Walk-in registration on event day is not available.
R 8	Team names must be unique and must not include offensive, trademarked, or misleading terms. The organising team reserves the right to reject or modify team names.
R 9	Team composition is locked at the time of registration. Merging with another team or adding/removing members after registration closes is not permitted.

4.3 Project Requirements

R 1 0	Each team must build a functional MCP server that implements at least two of the three MCP primitives: Tools, Resources, or Prompts.
R 1 1	All projects must be original work created during the hackathon (12:00 PM Day 1 to 12:00 PM Day 2). Pre-existing projects, coursework submissions, or clones of published MCP servers are not permitted.
R 1 2	Teams may use open-source libraries, publicly available datasets, and third-party APIs. All external components must be declared in the submission write-up. Undisclosed use of third-party code may result in disqualification.
R 1 3	Projects must be deployed and publicly accessible via a URL generated through the Nitrostack platform. Projects that are not live at the time of the submission deadline will not be evaluated.
R 1 4	Each team must select exactly one track at the time of registration. The track determines which judging pool the team competes in.
R 1 5	The MCP server must be built using the Nitrostack SDK or CLI. Teams wishing to use an alternative framework must seek written approval from the organising team before 06:00 AM on Day 2.

4.4 Submission Requirements

R 1 6	Submissions must be completed by 12:00 PM IST on 18 July 2026. No extensions will be granted under any circumstances. Prepare your submission well before the deadline.
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R 1 7	Teams must submit through TWO channels simultaneously: (a) the Nitrostack submission portal, and (b) the McpToTheMoon Discord server (#amrita-hackathon channel) and Reddit community. The McpToTheMoon submission is for community visibility and showcasing your work — it does not constitute the official judged submission.
R 1 8	The Nitrostack portal submission must include: (a) a deployed public URL of the MCP server, (b) a demo video of no more than 2 minutes, (c) a written description of 200–400 words covering the problem, solution, and potential impact, and (d) a link to a public source code repository (GitHub or GitLab).
R 1 9	The demo video must show the MCP server operating end-to-end, connected to at least one MCP-compatible client. Screen recordings with voiceover commentary are acceptable.
R 2 0	All team members' names and student roll numbers must be included in the submission form.

4.5 Conduct & Integrity

R 2 1	Plagiarism, code theft, or use of another team's work — in whole or in part — will result in the immediate disqualification of all teams involved.
R 2 2	AI coding assistants (GitHub Copilot, Claude, ChatGPT, Cursor, and similar tools) are permitted for implementation assistance. However, the problem definition, architectural design, and creative direction must originate from the team.
R 2 3	All participants must treat fellow participants, mentors, judges, and organisers with respect. Harassment, discrimination, or disruptive behaviour will result in removal from the event without appeal.
R 2 4	Teams must not attempt to access, disrupt, or interfere with any other team's deployed infrastructure or services.
R 2 5	Appeals against judging decisions must be submitted in writing to the Chief Organiser within 30 minutes of the winner announcement. Appeals will be reviewed by the Chief Organiser and one neutral faculty representative. Their decision is final.
R 2 6	All submitted projects remain the intellectual property of the team. By submitting, teams grant Wekan Enterprise Solutions, Nitrostack, and McpToTheMoon a non-exclusive, royalty-free right to feature the project in promotional and educational materials with full attribution to the team.

5. Themes & Industry Tracks

The hackathon is structured around six industry-aligned tracks. Each track represents a domain where Agentic AI and MCP servers can create measurable real-world impact. Teams must select exactly one track at registration. Judging criteria are identical across all tracks.

TRACK	THEME & FOCUS AREA	EXAMPLE MCP SERVER / APP IDEAS
 BFSI & FinTech	Banking, financial services, insurance — fraud detection, compliance automation, personal finance agents	<i>Loan eligibility assessment agent · Real-time fraud alert MCP server · Regulatory compliance checker (RBI/SEBI rules) · Portfolio rebalancing advisor · Insurance claim triage MCP</i>
 Manufacturing & Industry 4.0	Smart factories, supply chain, quality control, predictive maintenance using agentic AI	<i>Predictive maintenance agent (sensor data → work order) · Defect detection MCP with vision tools · Supply chain disruption monitor · Production scheduling optimiser MCP</i>
 IoT & Smart Infrastructure	Connected devices, smart cities, utilities, building management systems	<i>Smart grid demand-response agent · Building energy management MCP · Fleet tracking & route optimiser · Smart irrigation controller · Anomaly detection for industrial sensors</i>
 HealthTech & Life Sciences	Patient care, diagnostics, hospital operations, pharma research	<i>Clinical trial data aggregator MCP · Drug interaction checker agent · Appointment & bed management MCP · Medical literature summariser · Remote patient monitoring alert agent</i>
 Education & Research	Learning management, assessment, academic research, campus administration	<i>Adaptive assessment generator MCP · Research paper discovery & summariser · Campus timetable & resource booking agent · Student academic progress monitor · Plagiarism & citation checker MCP</i>
 Open Innovation	Any domain not listed above — legal, agriculture, logistics, governance, creator economy, cyber security	<i>Legal contract analyser MCP · Crop advisory agent (weather + soil data) · Last-mile logistics optimiser · Cyber threat intelligence MCP · Government scheme eligibility checker</i>

What makes a strong track entry?

The best submissions are not generic chatbots. They are purpose-built MCP servers that expose domain-specific tools and data sources. For example, an MCP server that connects an AI agent to live NSE/BSE market data and executes rule-based portfolio checks (BFSI), or one that ingests real-time sensor streams from IoT devices and triggers maintenance workflows (Manufacturing). Focus on agent functionality.

The Open Innovation track exists for teams whose idea genuinely does not fit the other five tracks. Teams selecting Open Innovation must clearly justify their domain choice in the submission write-up.

6. Judging, Evaluation & Panel

Projects are evaluated by a panel of three judges: one from each of the three partner organisations. Judges assess submissions independently using the weighted rubric below. Scores are aggregated to identify the top three teams per track.

Judging Panel

ORGANISATION	JUDGE ROLE	FOCUS AREA
Wekan Enterprise Solutions	Industry Judge	Evaluates real-world applicability, enterprise viability, and domain relevance of submissions
Nitrostack	Technical Judge	Evaluates MCP implementation quality, code architecture, deployment completeness, and Nitrostack platform usage
Amrita University	Academic Judge	Evaluates innovation, research potential, learning outcomes, and educational impact of the project

Evaluation Rubric

CRITERION	WHAT WE LOOK FOR	WEIGHT
Technical Quality	Clean MCP server implementation: correct use of Tools, Resources & Prompts; well-structured code; error handling; security considerations	25%
Innovation & Creativity	Originality of the idea; novel use of MCP primitives; unexpected or impactful combinations of tools and data sources	25%
Real-World Impact	Does it solve a genuine industry problem? Is the use case compelling, feasible, and scalable in the chosen track domain?	20%
Demo & Presentation	Quality of the 2-minute video; clarity of the deployed link; storytelling, pitch coherence, and communication of value	15%

CRITERION	WHAT WE LOOK FOR	WEIGHT
Completeness	Working deployment on Nitrostack; integration with at least one external data source or industry API; end-to-end functionality	10%
Community Visibility	Quality of the McpToTheMoon Discord and Reddit post — this is for showcasing work, not scored competitively	5%

Note on Community Visibility (5%)

The McpToTheMoon Discord and Reddit post is primarily a platform for showcasing your work to the global MCP builder community and gaining visibility as a developer. The 5% weight recognises the effort of communicating your project clearly to a public audience. It is not a popularity contest and community votes are not counted.

Judging Process

1. 06:01 PM: Automated checks run: deployment reachability, submission completeness, MCP primitive count.
2. 06:00–06:45 PM — Judges independently review demo videos and written descriptions.
3. 06:45–07:00 PM — Scores aggregated and winners confirmed by all three judges.
4. 07:00 PM — Closing Ceremony begins; winners announced.

7. Prizes, Awards, Budget & Opportunities

Prizes are awarded per track. With 6 tracks and 3 winners per track, 18 podium positions are available. The total prize budget is ₹1,44,000 . All prizes are denominated in round Indian Rupee (INR) amounts. Every registered team also receives Nitrostack platform credits to build and deploy their MCP servers throughout the event at no cost. All participants who submit a valid project receive a participation certificate.

AWARD	SCOPE	PRIZE MONEY	ADDITIONAL PERKS
 1st Place	Per Track — ₹12,000 per track	₹72,000 total	Trophy + Certificate + Nitrostack Pro License (1 yr) + Wekan Enterprise Credits + Nitrostack Platform Credits